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Degree: B.Tech. Semester: 5th - Course work
END-SEMESTER THEORY EXAMINATION, NOV-DEC,2024

Course Code: CGMTE02 Course Title: Probability & Statistics

Time: 03 Hours Max. Marks: 50

Note: - Attempt all the five questions. Missing data/ information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q 1	Attempt any 2 parts of the following		
1a	Find the probability of not getting a 3 or 7 or 11 total on either of two tosses of a pair of fair dice. Prove the results you use.	5	1
1b	Two cards are drawn from a well-shuffled ordinary deck of 52 cards. Find the probability that they are both aces if the first card is (i) replaced, (ii) not replaced.	5	1
1c	Prove Bayes' theorem. Box I contains 3 red and 2 blue marbles while Box II contains 2 red and 8 blue marbles. A fair coin is tossed. If the coin turns up heads, a marble is chosen from Box I; if it turns up tails, a marble is chosen from Box II. Suppose that a red marble was chosen. What is the probability that Box I was chosen (i.e., the coin turned up heads)?	5	1
Q 2	Attempt any 2 parts of the following		
2a	Find the mean and variance of the uniform distribution on the interval (a, b).	5	2
2b	Find the mean and variance of the gamma distribution with parameters a, b.	5	2
2c	Find the mean and variance of the beta distribution.	5	2
Q 3	Attempt any 2 parts of the following		
3a	Show that (i) the moment generating function for a Cauchy distributed random variable X does not exist but that (ii) the characteristic function does exist.	5	3
3b	Find the moment generating function of binomial random variable with parameters (n, p). Find variance using MGF.	5	3

3c	State central limit theorem and weak law of large numbers.	5	3
Q 4	Attempt any 2 parts of the following		
4a	A population consists of the five numbers 2, 3, 6, 8, 11. Consider all possible samples of size two which can be drawn with replacement from this population. Find (i) the mean of the population, (ii) the standard deviation of the population, (iii) the mean of the sampling distribution of means.(iv) the standard deviation of the sampling distribution of means, i.e., the standard error of means.	5	4
4b	Solve 4a in case sampling is without replacement.	5	4
4c	Find the probability that in 120 tosses of a fair coin (i) between 40% and 60% will be heads, (ii) 0.625 or more will be heads.	5	4
Q 5	Attempt any 2 parts of the following		
5a	State theorems relating t, F and chi-square distributions.	5	5
5b	Write the moment generating function of t-distribution. Compute its mean and variance.	5	5
5c	A sample of 10 measurements of the diameter of a sphere gave a mean x =4.38 inches and a standard deviation s = 0.06 inch. Find (i) 95%, (ii) 99% confidence limits for the actual diameter. (given t(0.975) = 2.26, t(0.995) =3.25 with 9 degrees of freedom.)	5	5